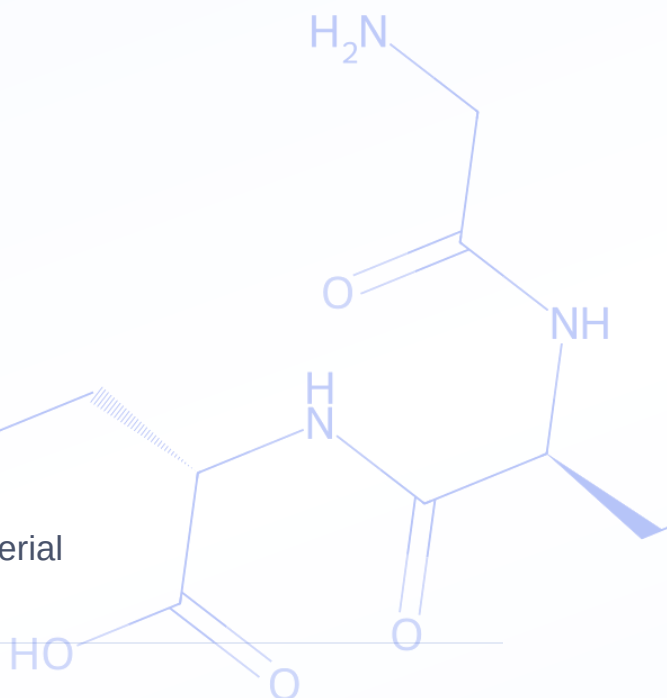


● RESEARCH GRADE · 99%+ PURITY · USA-BASED

RESEARCH REFERENCE & SAFETY DOCUMENTATION

GHK-Cu

Copper tripeptide-1 · Research-grade reference material



CAS (COMPLEX)

89030-95-5

MW (COMPLEX)

≈403.9 g/mol

COMPOSITION

Cu(II)-GHK 1:1

CLASS

Copper-peptide complex

Identity & Composition

GHK-Cu is the copper(II) complex of the tripeptide glycyl-L-histidyl-L-lysine (Gly-His-Lys), bound 1:1. First isolated from human plasma (Pickart, 1973); endogenous GHK declines with age. Supplied lyophilized, research grade.

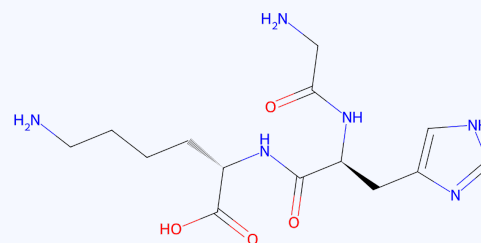
Product name	GHK-Cu (Copper Tripeptide-1)
Composition	Cu(II) complex of Gly-His-Lys, 1:1
CAS (complex)	89030-95-5
CAS (free GHK)	49557-75-7
MW	≈403.9 (complex) / 340.38 (free)
PubChem CID	71587328 (complex) / 73587 (free)
Appearance	Blue to blue-violet lyophilized powder
Solubility	Soluble in water
Identified use	In-vitro research & development only

Gly

His

Lys

Tripeptide backbone: glycyl-L-histidyl-L-lysine, complexed with Cu(II)



Molecular structure. Glycyl-L-histidyl-L-lysine tripeptide (free GHK, C₁₄H₂₄N₆O₄, 340.38 g/mol). Supplied as the 1:1 copper(II) complex (GHK-Cu, ≈403.9 g/mol); the copper center is integral to activity.

Copper complex. Unlike simple peptides, GHK-Cu contains chelated copper(II). This affects handling, ecological and transport considerations — see the Safety Data Sheet section for copper-specific guidance.

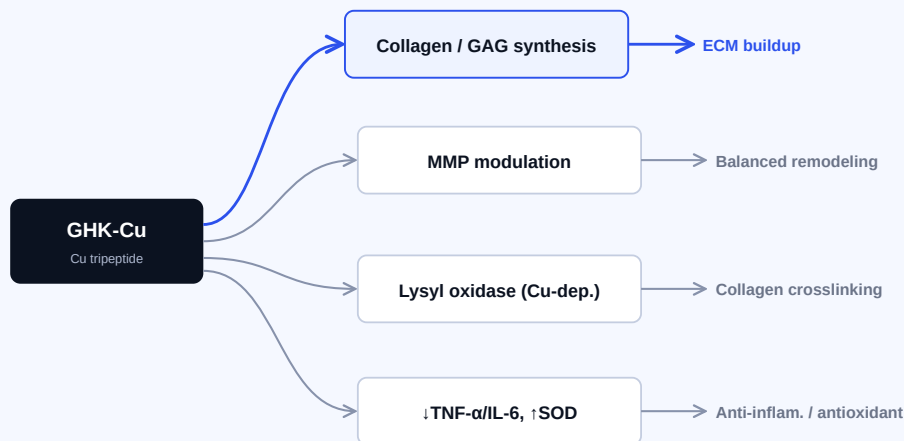
Research Background

GHK-Cu has been studied for decades in skin-biology, wound-healing and extracellular-matrix (ECM) research. The observation that circulating GHK declines with age motivated much of the interest in exogenous GHK-Cu as a research tool in connective-tissue and regenerative biology.

Mechanisms studied in preclinical models

Findings below derive from in-vitro and animal models (with some early/limited human topical data). Not established systemic effects in humans.

- **Collagen & GAG synthesis.** Dose-dependent stimulation of collagen (types I & III) and glycosaminoglycan synthesis in cultured dermal fibroblasts (Maquart et al., 1988).
- **“Balanced” matrix remodeling.** Modulation of matrix metalloproteinases (e.g., MMP-2) — clearing damaged matrix while supporting new synthesis.
- **Copper-dependent enzyme activity.** Influence on lysyl oxidase (collagen/elastin crosslinking); the copper center is integral.
- **Anti-inflammatory / antioxidant signaling.** Downregulation of TNF- α and IL-6; modulation of antioxidant enzymes (e.g., SOD) in wound models.



Proposed pathways investigated in preclinical (in-vitro / animal) models. Schematic only; not a claim of established human effect.

What the Literature Supports — and What It Doesn't

GHK-Cu is among the most extensively studied tripeptides in skin-biology and ECM research. The collagen-synthesis and remodeling findings in fibroblast and wound models are among the most replicated in the area; human-relevant evidence is strongest for topical use.

Evidence at a glance

PRECLINICAL (IN-VITRO/ANIMAL)

Extensive

HUMAN — TOPICAL

Moderate

HUMAN — SYSTEMIC/INJECTABLE

Limited

REGULATORY

Not an approved drug

- Collagen/ECM findings in fibroblast & wound models are among the **most replicated** in this area (since the 1980s).
- The best human-relevant evidence concerns **topical skin/wound** applications; some recent work uses objective measures (e.g., ultrasound dermal collagen density).
- **Injectable/systemic use is not well supported**, and human safety data for injectable GHK-Cu is limited.

Appropriate characterization: a well-studied copper-tripeptide research tool in skin/ECM biology, with strong preclinical mechanistic data and limited, mostly-topical human evidence.

Regulatory & status notes

US FDA	Appears in cosmetic contexts; injectable/compounded GHK-Cu flagged by FDA over limited human safety data. Not an approved drug for systemic therapeutic use.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

1. Maquart FX, et al. Stimulation of collagen synthesis in fibroblast cultures by the tripeptide-copper complex glycyl-L-histidyl-L-lysine-Cu²⁺. FEBS Lett. 1988. PMID 3169264.
2. Pickart L, Margolina A. Regenerative and protective actions of the GHK-Cu peptide (review).
3. Exploring the Role of Tripeptides in Wound Healing and Skin Regeneration: A Comprehensive Review. Int J Med Sci. 2025;22:4175.

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	GHK-Cu (Copper Tripeptide-1)
CAS (complex/free)	89030-95-5 / 49557-75-7
MW	≈403.9 (complex) / 340.38 (free)
Composition	Cu(II) complex of Gly-His-Lys, 1:1
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use
Supplier	
Emergency	

2 Hazard Identification

Not classified based on available data for the intact complex; limited data. Copper compounds as a class can carry irritation and, for some, aquatic-toxicity considerations (see §11–12) — a qualified reviewer should confirm whether a copper-based classification applies. **Precautionary:** P261, P264, **P273 (avoid release to environment)**, P280, P501. Handle as a bioactive copper-peptide complex of incompletely characterized toxicity.

3 Composition

Copper(II) complex of glycyl-L-histidyl-L-lysine (1:1), >98% research grade. Contains chelated copper(II). Residual reagents/solvents/counter-ions may be present. See lot COA for impurity profile.

4 First-Aid Measures

Eyes: rinse with water. **Skin:** wash with soap and water. **Inhalation:** fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Ingestion of copper compounds may cause GI irritation. Treat symptomatically.

5 Fire-Fighting

Media: CO₂, dry chemical, foam, water spray. Combustion may produce CO, CO₂, NO_x and copper-containing particulate. Use SCBA and full protective gear.

6 Accidental Release

Avoid dust; ventilate; use PPE per §8. **Do not allow entry to drains/waterways — copper compounds may be harmful to aquatic life.** Collect in a closed labeled container for disposal per §13.

7 Handling & Storage

Handle as bioactive copper-peptide complex; minimize exposure; protect from light. Store lyophilized solid tightly closed, dry, cool; -20 °C long-term, 2–8 °C short-term. Solutions are light/oxidation-sensitive — aliquot, store frozen, protect from light. Observe COA retest date.

8 Exposure Controls / PPE

No substance-specific OEL; general copper-dust/fume limits may inform handling. Control to ALARA. Fume hood for dry powder. Nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator where controls insufficient.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	Blue to blue-violet powder
Odor	Odorless
Solubility	Water
MW	≈403.9 g/mol (complex)
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable under normal storage; protect from light. Avoid heat, moisture, UV, strong reducing conditions. Incompatible with strong reducing agents, strong acids/bases, and strong chelators that could displace copper. Decomposition: CO, CO₂, NO_x, copper oxides. No hazardous polymerization.

11 Toxicological Information

Intact complex not fully characterized. No substance-specific LD₅₀/LC₅₀. Copper compounds can be irritating — avoid contact. Sensitization: cannot be ruled out. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: no substance-specific data; not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

Copper-containing compounds can be toxic to aquatic organisms. No substance-specific data for the complex; as a precaution treat as potentially hazardous to the aquatic environment and avoid release to surface water, soil, drains or sewers. A qualified reviewer should confirm whether GHS aquatic-toxicity classification applies.

13 Disposal

Dispose per applicable regulations; copper content may affect waste classification. US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not expected to be regulated under DOT/IATA/IMDG based on current classification, **but confirm** given copper content and potential aquatic-toxicity/marine-pollutant considerations. Shipper must independently verify.

15 Regulatory Information

US TSCA: may qualify for R&D exemption (40 CFR 720.36). OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption. Not a legal determination of regulatory status.

16 Other Information

Revision [DATE] · Version 1.0. Independent professional review recommended prior to use — note copper-specific classification items above. Sources: PubChem; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8. Provided without warranty. For scientific R&D use only.

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